# Fine-Tuning Egocentric Visual Encoders on Exocentric Data for Robot Manipulation

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***Abstract*—Visual representations pretrained on large-scale human video have emerged as a promising approach to reducing the data burden in robot manipulation learning. R3M and VIP, both trained on the egocentric Ego4D dataset, achieve strong performance in low-data behavioral cloning settings, yet share a fundamental limitation: their pretraining data is captured entirely from a first-person human perspective, while robot manipulation cameras are typically fixed third-person observers. This viewpoint mismatch may limit the transferability of learned features to robot deployment settings. We reproduce the behavioral cloning evaluations of both R3M and VIP on five Meta-World manipulation tasks across three demonstration sizes, three camera viewpoints, and three random seeds, confirming baseline performance of 72.8% and 72.0% mean success rate respectively. We then investigate whether fine-tuning on exocentric robot manipulation data can reduce this domain gap, using a subset of the DROID dataset, a large-scale collection of 76,000 real-world robot demonstrations captured from diverse third-person cameras. Contrary to our hypothesis, both full backbone and frozen backbone fine-tuning strategies degraded performance by approximately 11 percentage points for both models. The near-identical results across both fine-tuning strategies rule out catastrophic forgetting as the primary cause of degradation. The remaining explanations, including morphological mismatch between the human hands depicted in Ego4D and the robot grippers in DROID, and visual domain shift between DROID's real-world environments and MetaWorld's simulation, are discussed but remain open questions for future investigation. These findings suggest that exocentric fine-tuning requires careful alignment between the fine-tuning data distribution and the downstream evaluation environment, and that not all third-person robot data is equally suited for improving egocentric-pretrained representations.**

## I. Introduction

Modern robotic systems have achieved impressive performance in narrowly defined tasks, often excelling in structured environments where objectives and conditions are carefully controlled. However, these systems remain highly specialized, as their capabilities are largely shaped by the specific datasets on which they are trained. Although efforts have been made to aggregate larger robotic datasets, they remain limited in scope relative to the wide range of tasks expected of a generalist robot. Achieving broader adaptability requires significantly more diverse and scalable training data. This raises a fundamental question: how can robotics access the volume and diversity of data necessary to move beyond task-specific systems toward generalist robotic intelligence?

One promising answer lies in Learning from Video (LfV), which leverages large-scale, freely available internet video to pretrain visual representations that can later be transferred to robot learning with minimal task-specific data. Rather than collecting expensive robot demonstrations from scratch for every new task, LfV methods extract generalizable visual structure from human behavior observed in the wild, dramatically reducing the data burden on downstream robot learning. Among the most successful instantiations of this approach are R3M [1] and VIP [2], both of which pretrain ResNet-50 encoders on the large-scale Ego4D human video dataset [3] and demonstrate strong performance in low-data imitation learning settings for robot manipulation.

Despite their effectiveness, both R3M and VIP share a fundamental characteristic of their pretraining data: Ego4D is an egocentric dataset, recorded entirely from first-person (head-mounted) cameras worn by human participants. Robot manipulation setups, by contrast, typically observe the scene from fixed third-person cameras. This viewpoint mismatch introduces a domain gap between what the encoders learned to represent and the visual inputs they receive during robot deployment. Whether this egocentric pretraining bias limits downstream performance, and whether it can be addressed through targeted fine-tuning on exocentric data, remains an open and underexplored question.

This paper investigates that question directly. We reproduce the behavioral cloning evaluations of both R3M and VIP on the MetaWorld manipulation benchmark, then fine-tune both encoders on a subset of the DROID dataset [4], a large-scale collection of real-world robot manipulation video captured from third-person cameras across diverse environments. DROID was selected as a practical alternative after access to the originally planned Ego-Exo4D dataset [5] could not be obtained within the project timeline. By comparing baseline and fine-tuned encoder performance under identical evaluation conditions, we aim to assess whether fine-tuning on exocentric robot manipulation data can meaningfully reduce the viewpoint domain gap and improve downstream manipulation performance.

## II. Background

### A. The Data Bottleneck in Robot Learning

Imitation learning in robotics is constrained by small, task-specific, and costly datasets [6]. Collecting large-scale robot demonstration data is inherently challenging: robots require extensive data to improve, yet their limited capabilities restrict the amount of meaningful data they can gather, creating a compounding bottleneck [7]. Although human teleoperation can be used to generate demonstrations, it is costly and difficult to scale [7]. Simulated environments offer a means of scaling data collection but suffer from the sim-to-real gap, as simulations cannot perfectly replicate real-world physics and environmental complexity [8].

Behavioral cloning (BC) is the dominant approach to imitation learning in this setting. Given a set of expert demonstrations, BC trains a policy to directly map observations to actions by minimizing the supervised regression loss between predicted and expert actions [9]. While conceptually simple, BC is highly sensitive to the amount and quality of demonstration data available: in the low-data regime (fewer than 25 demonstrations), performance degrades significantly unless the visual observations are encoded by a rich, transferable representation [1]. The quality of the visual encoder is therefore central to whether BC can succeed in practical, data-scarce robot learning settings.

### B. Learning from Video as a Scalable Alternative

In contrast to robot-specific datasets, the internet contains vast quantities of freely available video that is diverse in tasks, environments, and contextual interactions [7]. Learning from Internet Video (LfV) refers to harnessing this large-scale data, together with advances in deep learning, to pretrain visual representations that capture generalizable structure about physical interaction and task progress. The core intuition is that humans and robots perform structurally similar manipulation tasks, and that representations trained on human video should transfer to robotic settings even without explicit robot data.

LfV is not a single algorithmic technique but a multi-stage pipeline. The first stage involves pretraining a visual encoder on large-scale video using self-supervised objectives that extract temporal structure, semantic content, or goal-directed dynamics from raw footage [7]. The resulting frozen encoder is then used as the perception module for a downstream policy, which is trained on a small number of robot demonstrations using behavioral cloning. By decoupling representation learning from policy learning, LfV allows the expensive data collection to happen once at the pretraining stage, with only a small task-specific fine-tuning cost thereafter [1].

The primary benefit of this approach is dramatically improved sample efficiency. When equipped with a well-pretrained visual encoder, behavioral cloning policies can achieve strong performance with as few as five demonstrations on tasks that would otherwise require hundreds [1]. This makes LfV particularly attractive for real-world robot deployment where collecting large demonstration datasets is impractical.

LfV does, however, introduce its own challenges. Internet video lacks explicit action labels, so representations must be learned from visual structure alone without access to the control signals that generated observed behavior [7]. There is also an embodiment mismatch between human and robot morphology, and a distribution shift between the environments depicted in internet video and real robot workspaces. Careful design of the pretraining objective is therefore critical to ensuring that the learned representations are useful for robot control rather than merely good at reconstructing video.

### C. R3M: Representation Learning via Time Contrast and Language

R3M [1] is a visual representation pretrained on the Ego4D dataset [3] using a combination of three objectives. The first is time-contrastive learning, which trains the encoder to produce embeddings such that frames closer in time are more similar than frames far apart or from different videos. This encourages the representation to capture the temporal dynamics of task execution. The second objective is video-language alignment, which trains a language prediction module on top of the encoder to predict the natural language narration associated with each video clip. This encourages the representation to be semantically meaningful, capturing features relevant to describing actions in language. The third objective is L1 sparsity regularization, which encourages compact representations that are less prone to overfitting in low-data imitation learning. The combined objective is:

$$\mathcal{L}_{\text{R3M}} = \lambda_1 \mathcal{L}_{\text{tcn}} + \lambda_2 \mathcal{L}_{\text{lang}} + \lambda_3 \|F_\phi(I)\|_1 + \lambda_4 \|F_\phi(I)\|_2 \quad (1)$$

where $F_\phi$ is the ResNet-50 encoder, $\mathcal{L}_{\text{tcn}}$ is the time-contrastive loss, and $\mathcal{L}_{\text{lang}}$ is the video-language alignment loss. The resulting 2048-dimensional embeddings are used as frozen visual features for downstream behavioral cloning. R3M achieves approximately 62% average success rate across 12 manipulation tasks in the low-data imitation learning setting, outperforming all prior visual representations including CLIP and MoCo by more than 10% on average [1].

### D. VIP: Value-Implicit Pretraining

VIP [2] builds on the same pretraining paradigm as R3M but replaces the time-contrastive and language alignment objectives with a fully self-supervised value-implicit objective derived from dual reinforcement learning. Rather than treating representation learning as a vision problem, VIP frames it as solving a goal-conditioned offline RL problem over human video, where the goal of each video clip is its final frame. The resulting objective encourages the encoder to learn embeddings whose distances serve as a smooth, monotonically decreasing reward signal as the task progresses toward the goal. Setting the value function as the negative L2 embedding distance, $V^*(\phi(o), \phi(g)) := -\|\phi(o) - \phi(g)\|_2$, yields the final VIP training objective (Equation 6 in [2]):

$$\mathcal{L}(\phi) = \mathbb{E}_{p(g)}\Big[(1-\gamma)\,\mathbb{E}_{\mu_0(o;g)}\big[\|\phi(o)-\phi(g)\|_2\big] + \log \mathbb{E}_{(o,o';g)\sim\mathcal{D}}\big[\exp\big(\|\phi(o)-\phi(g)\|_2 - \tilde{\delta}_g(o) - \gamma\|\phi(o')-\phi(g)\|_2\big)\big]\Big] \quad (2)$$

where $\phi$ is the visual encoder, $o$ and $o'$ are consecutive observations, $g$ is the goal frame (final frame of the sub-trajectory), $\gamma$ is a discount factor, $\mu_0(o;g)$ is the goal-conditioned initial observation distribution, and $\tilde{\delta}_g(o) = \mathbf{1}[o = g] - 1$ is a sparse reward that is zero only when the observation equals the goal and $-1$ otherwise. Crucially, VIP requires no language annotations or action labels, making it fully self-supervised. Like R3M, VIP uses a ResNet-50 backbone pretrained on Ego4D, but produces 1024-dimensional embeddings. In visual imitation learning evaluations, VIP achieves comparable performance to R3M, with both methods significantly outperforming prior representations such as CLIP and ImageNet-supervised features [2].

### *E. The Egocentric Pretraining Bias*

A characteristic shared by both R3M and VIP is their exclusive reliance on Ego4D for pretraining. Ego4D consists entirely of egocentric video recorded from head-mounted cameras worn by human participants engaged in everyday activities [3]. While this provides a large and diverse source of human-object interaction data, it introduces a systematic viewpoint bias: all pretraining observations are from a first-person perspective, whereas robot manipulation cameras are almost universally positioned as fixed third-person observers of the workspace.

This viewpoint mismatch means that the low-level and mid-level visual features learned during pretraining may not optimally represent the spatial relationships, object configurations, and motion patterns that are most informative from a third-person perspective. The Ego-Exo4D dataset [5] was originally identified as an ideal source of exocentric fine-tuning data for this purpose, as it provides synchronized egocentric and exocentric recordings of the same human activities, making it a direct bridge between the Ego4D pretraining distribution and third-person robot deployment cameras. However, access to Ego-Exo4D could not be obtained within the project timeline, and the DROID dataset [4] was used instead as a practical alternative. Whether fine-tuning on exocentric data from either source can measurably reduce the viewpoint domain gap and improve downstream manipulation performance is the central empirical question of this work.

## III. Related Work

### *A. Visual Representations for Robot Manipulation*

A growing body of work has explored pre-trained visual representations as frozen perception modules for robot learning, motivated by the observation that training visual encoders from scratch in the low-data regime yields poor performance [1]. The central question across these works is what pretraining data and objective best produces representations that transfer to robot manipulation.

CLIP [10] is among the most widely evaluated baselines in this space. Trained on internet-scale image-text pairs via contrastive learning, CLIP produces semantically rich visual features that have shown utility in navigation and some manipulation tasks. However, because CLIP is trained on static images with no temporal structure, it lacks the ability to capture task progress or sequential dynamics, which are critical for manipulation. Both R3M and VIP report that CLIP underperforms representations trained specifically on human interaction video by more than 10% on average across manipulation tasks [1], [2].

MVP [11] takes a different approach, pretraining a masked autoencoder (MAE) on a large collection of internet images and egocentric video. Unlike R3M and VIP, MVP does not rely on temporal contrastive or value-based objectives; instead it uses masked image reconstruction as the sole pretraining signal. MVP demonstrates that self-supervised pretraining on diverse real-world images can outperform supervised ImageNet pretraining by up to 80% on motor control tasks, and that egocentric and in-the-wild internet video both contribute to stronger representations than ImageNet images alone. The success of MVP reinforces the importance of pretraining data distribution, though its reconstruction-based objective does not explicitly encode temporal dynamics or goal-directed progress in the way that R3M and VIP do.

VC-1 [12] scales the masked autoencoder approach further, training a Vision Transformer (ViT) on over 4,000 hours of egocentric video from seven sources combined with ImageNet, totaling over 5.6 million images. VC-1 is evaluated on CortexBench, a suite of 17 tasks spanning locomotion, navigation, and dexterous and mobile manipulation, making it the most comprehensive benchmark comparison in the pretrained representation literature to date. A key finding of VC-1 is that no single pretrained representation is universally dominant across all tasks, and that task-specific or domain-specific adaptation of VC-1 yields substantial gains beyond the frozen encoder baseline. This finding motivates the fine-tuning approach taken in the present work: rather than relying on a frozen encoder pretrained on generic egocentric data, we investigate whether targeted adaptation to exocentric data can improve performance on the specific viewpoint distribution encountered during robot deployment.

Taken together, MVP and VC-1 establish that the choice of pretraining data distribution matters significantly, and that egocentric human video is a particularly strong source of features for manipulation. However, neither work explicitly addresses the viewpoint mismatch between egocentric pretraining data and exocentric robot deployment cameras, which is the gap this work targets.

### *B. Vision-Language-Action Models*

Vision-Language-Action (VLA) models represent a distinct and more recent direction in robot learning, integrating visual

perception, natural language understanding, and action generation into unified end-to-end architectures [13]. Rather than using a pretrained visual encoder as a frozen module for a separate policy, VLA models treat the entire perception-to-action pipeline as a single model trained jointly on large-scale robot data.

RT-1 [14] was among the first large-scale VLA models, demonstrating that a robotics transformer trained on over 130,000 real robot demonstrations could generalize to a wide range of manipulation tasks. RT-2 [15] extended this further by fine-tuning a large vision-language model on robot data, enabling the transfer of web-scale semantic knowledge to robot control. More recently, $\pi_0$ [16] introduced a flow-matching action expert combined with a pre-trained vision-language backbone, achieving strong performance on complex long-horizon manipulation tasks. These models represent a significant scaling up of both data and compute relative to representation-based approaches like R3M and VIP.

While VLA models have shown impressive results, they require substantially more data, compute, and infrastructure than representation-based methods. R3M and VIP, by contrast, can be deployed with as few as five demonstrations and a small MLP policy head, making them far more accessible for low-resource settings. The relationship between these two paradigms is an open question: whether the egocentric pre-training improvements explored in this work might also benefit the visual backbone components of VLA models is a natural direction for future investigation.

### C. Exocentric and Cross-Viewpoint Learning

A small but relevant body of work has investigated the gap between egocentric and exocentric visual representations directly. The Ego-Exo4D dataset [5] was introduced specifically to enable research on understanding skilled human activity from both first- and third-person perspectives simultaneously, providing synchronized egocentric and exocentric recordings of the same activities across a range of physical tasks. While the dataset was not designed specifically for robot learning, its manipulation-relevant scenario categories (including cooking and tool use) make it a natural source of exocentric pretraining data for encoders like R3M and VIP that were trained exclusively on egocentric footage.

Prior work in cross-viewpoint imitation learning has noted that the distribution shift between observation viewpoints can significantly affect policy performance [1], and that representations trained on egocentric data do not always transfer cleanly to exocentric robot cameras [7]. This work originally intended to use Ego-Exo4D as the fine-tuning source, given its direct correspondence between egocentric and exocentric views of the same activities. Due to data access constraints, the DROID dataset [4] was used instead. To our knowledge, no prior work has directly investigated fine-tuning R3M or VIP on exocentric robot manipulation data as a means of reducing the viewpoint gap, making this an open and underexplored direction in the intersection of cross-viewpoint representation learning and robot manipulation.

## IV. Methodology

### A. Overview

This work investigates whether fine-tuning the visual representations of R3M and VIP on exocentric (third-person) video data can improve their downstream performance on robot manipulation tasks. The methodology proceeds in three phases: (1) baseline reproduction of the original R3M and VIP behavioral cloning evaluations on MetaWorld, (2) fine-tuning both encoders on a subset of the DROID robot manipulation dataset using third-person camera footage, and (3) re-evaluation of the fine-tuned models under identical conditions to enable direct comparison.

### B. Representations Under Investigation

Two pre-trained visual representations are evaluated. R3M [1] is a ResNet-50 encoder pre-trained on the Ego4D dataset [3] using a combination of time-contrastive learning, video-language alignment, and L1 sparsity regularization. Its training objective encourages the representation to capture temporal dynamics and semantically relevant features from first-person human video. VIP [2] is also a ResNet-50 encoder pre-trained on Ego4D, but uses a fully self-supervised value-implicit objective derived from dual reinforcement learning, requiring no language annotations or action labels. Both models produce frozen visual embeddings that are concatenated with the robot's proprioceptive state and passed to a downstream policy network. A key distinction between the two is embedding dimensionality: R3M produces 2048-dimensional embeddings while VIP produces 1024-dimensional embeddings, which required separate demonstration data pipelines for each model.

### C. Simulation Environment and Tasks

All evaluations are conducted in the MetaWorld simulation benchmark [17]. Five manipulation tasks are used, matching the subset evaluated in the original R3M paper: Button Press, Assembly (ring on peg), Bin Picking, Drawer Opening, and Hammer. All tasks use the Sawyer robot arm with a 4-dimensional action space consisting of end-effector displacement and gripper open/close. Object positions are randomized between episodes.

The evaluation protocol follows the original paper: a 500-step rollout horizon, three camera viewpoints (front, side, and top-down), and three demonstration dataset sizes (5, 10, and 25 episodes). Each condition is evaluated over 20 rollout episodes and run with three random seeds. Results are reported as mean success rate with standard deviation, averaged across seeds.

### D. Demonstration Data Collection

Expert demonstration data is generated using scripted and heuristic policies within MetaWorld. For most tasks, MetaWorld's built-in scripted policies are used to collect successful trajectories. The Assembly task required a custom solution: the built-in policy was found to be incompatible with the v3 environment, as the peg position is not exposed in the observation vector. A six-phase inverse kinematics heuristic

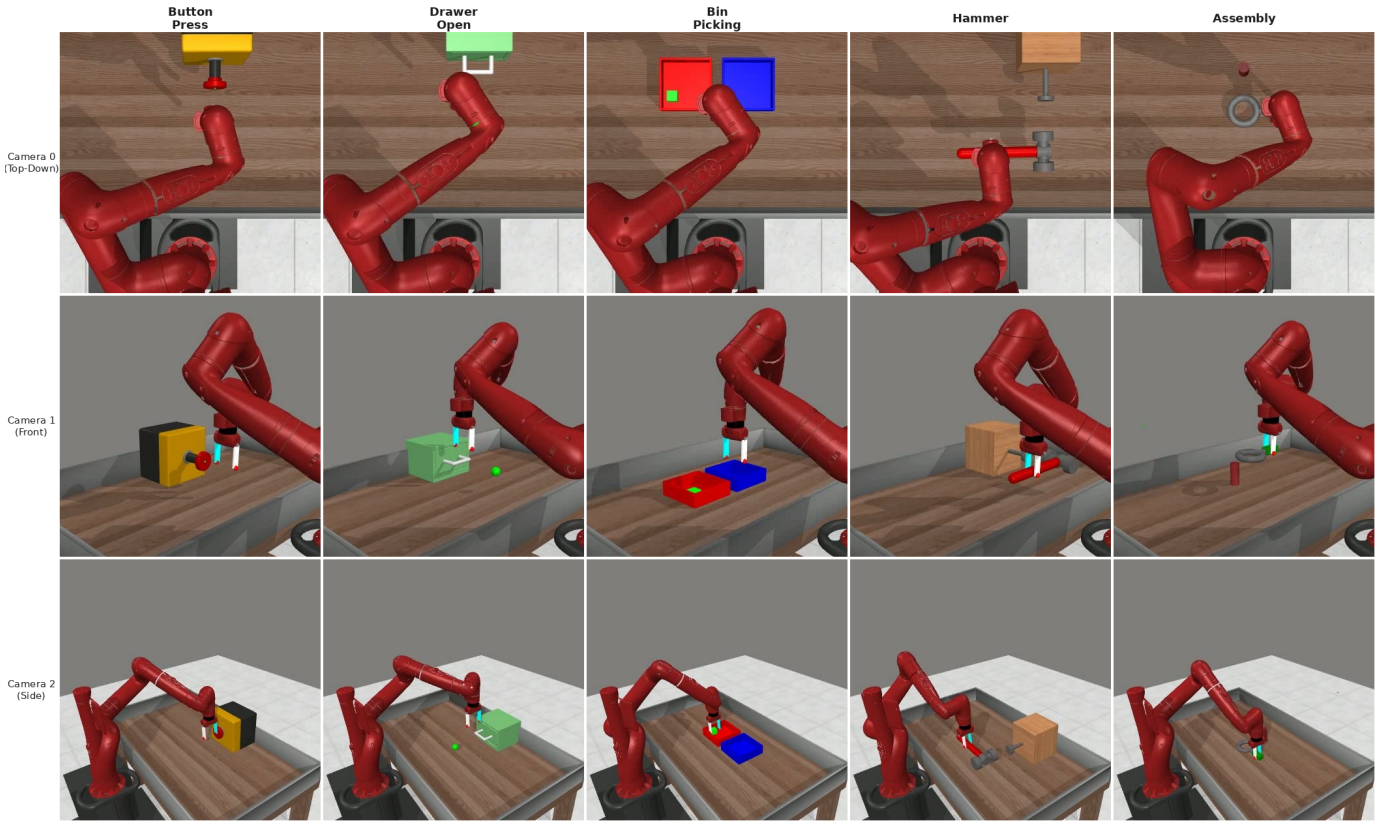



Fig. 1: MetaWorld evaluation tasks and camera viewpoints. Each column corresponds to one of the five manipulation tasks evaluated in this work. Each row shows the same task from a different fixed camera: top-down (Camera 0), front (Camera 1), and side (Camera 2). The viewpoint variation across rows illustrates the egocentric-to-exocentric domain gap that motivates this work.

was developed that reads the peg position directly from the MuJoCo physics engine [18] to construct valid assembly trajectories. Only successful episodes, as determined by the environment's binary success signal, are retained for training. Due to differences in embedding dimensionality between R3M and VIP, demonstration data is encoded and stored separately for each model.

Obtaining valid demonstration data proved to be the most significant practical challenge of this work. Franka Kitchen and Adroit evaluations were attempted but ultimately excluded: available Franka Kitchen demonstration data contained insufficiently varied action sequences that could not be filtered into distinct task demonstrations, and Adroit demonstration data exhibited coordinate and seeding mismatches incompatible with the current environment version. RoboMimic was also evaluated as an alternative source of demonstration data but could not be successfully integrated. MetaWorld was the only environment for which fully compatible demonstration data could be obtained or generated.

### *E. Behavioral Cloning Policy*

Following the protocol of both the R3M and VIP papers, a two-layer MLP with hidden sizes $[256, 256]$ preceded by a BatchNorm layer is used as the downstream policy. The policy takes as input the concatenation of the frozen visual embedding and the robot's proprioceptive state, and outputs a 4-dimensional action. The policy is trained with a mean squared error behavioral cloning loss:

$$\mathcal{L}_{\text{BC}} = \|a_t - \pi([z_t, p_t])\|_2^2 \tag{3}$$

where $a_t$ is the expert action, $z_t = F_\phi(I_t)$ is the frozen visual embedding of the current observation with $I_t$ denoting the RGB image frame at timestep $t$ and $F_\phi$ denoting the pretrained visual encoder with frozen parameters $\phi$, and $p_t$ is the robot proprioceptive state. The policy is optimized using Adam [19] with a learning rate of $10^{-3}$ and batch size of 32.

Training follows the R3M paper's fixed gradient step protocol of 20,000 steps regardless of demonstration dataset size, with random sampling with replacement from the collected transitions at each step. This differs from epoch-based training in that it provides a consistent training budget across all demonstration sizes, enabling fair comparison across the 5, 10, and 25 demonstration conditions.

### *F. Implementation Challenges*

Several implementation challenges were encountered during the reproduction of the original experiments.

**Evaluation environment coverage.** As noted above, only MetaWorld evaluations could be completed due to incompatibilities with available demonstration data for Franka Kitchen and Adroit. This represents a limitation relative to the original papers, which reported results across all three environments.

**Embedding dimension incompatibility.** R3M and VIP produce embeddings of different dimensionalities (2048 and 1024, respectively), which prevented a shared demonstration data pipeline. Each model requires its own encoded dataset, effectively doubling the data preparation overhead and preventing direct sharing of pre-encoded transitions between the two evaluation pipelines.

**Python 3.12 compatibility.** VIP's Hydra-based configuration system exhibited a compatibility issue with Python 3.12, in which several dataclass fields in the Hydra core configuration module used mutable default values disallowed by the language version. This was resolved by patching several dataclass fields in the Hydra core configuration module to use `field(default_factory=...)` entries.[1]

### G. Exocentric Fine-Tuning

Both R3M and VIP were originally pre-trained exclusively on Ego4D [3], a large-scale egocentric video dataset recorded from a first-person (head-mounted camera) perspective. Robot manipulation setups, by contrast, typically observe the scene from a fixed third-person camera. This viewpoint mismatch represents a domain gap that may limit the transferability of features learned during pre-training. To investigate this, both encoders are fine-tuned on a subset of the DROID dataset [4].

**Dataset.** DROID (Distributed Robot Interaction Dataset) is a large-scale, in-the-wild robot manipulation dataset comprising 76,000 human-teleoperated demonstration trajectories totaling 350 hours of interaction, collected across 564 unique real-world scenes, 86 distinct manipulation tasks, and 52 buildings across three continents over 12 months [4]. All data was collected using the same Franka Panda robot arm hardware. Each episode contains three synchronized third-person RGB camera streams, camera calibration data, depth information, and natural language instructions. DROID covers 1,417 distinct camera viewpoints, making it one of the most viewpoint-diverse robot manipulation datasets available. The full dataset totals 1.7TB across 2,050 shards and is released under the CC-BY 4.0 license.

Due to local storage constraints, a subset of approximately 6,000 episodes ($\sim$8% of the full dataset) was used for fine-tuning. Only the exterior third-person camera stream (`exterior_image_1_left`) was used; the wrist-mounted camera was excluded. DROID was selected as a practical alternative after access to the originally planned Ego-Exo4D dataset [5] could not be obtained. Its large scale, third-person viewpoint coverage, language annotations, and open license made it a suitable substitute. Additionally, as actual robot demonstration data, DROID represents a directly relevant exocentric distribution for the downstream manipulation tasks being evaluated.

**Fine-tuning strategy.** Two fine-tuning strategies were evaluated to assess the role of catastrophic forgetting. In the first, the full ResNet-50 encoder was updated, including the convolutional backbone and the final projection head. In the second, the convolutional backbone was frozen and only the final projection head was updated, eliminating the risk of overwriting features learned during Ego4D pretraining. Both strategies preserved the original R3M training objectives: the time-contrastive loss was applied across temporally ordered frames sampled from each DROID episode using the `is_first` and `is_last` flags to ensure clean episode boundaries. Video-language alignment was included conditionally using the DistilBERT language model [20] to encode DROID's natural language instructions when available. L1 and L2 sparsity regularization were preserved with the original hyperparameter values ($\lambda_3 = \lambda_4 = 10^{-5}$). Both strategies used the Adam optimizer with a learning rate of $3 \times 10^{-5}$ for 5 epochs.

**Future directions.** Partial backbone fine-tuning (unfreezing only the final residual block), fine-tuning on Ego-Exo4D exocentric video once access is obtained, and fine-tuning on a larger or more task-relevant subset of DROID all represent natural extensions of this work.

## V. Evaluation

### A. Baseline Performance

We first evaluate the unmodified R3M and VIP encoders on the five MetaWorld tasks across all three demonstration sizes, three camera viewpoints, and three random seeds. Results are reported in Table I and visualized in Fig. 2.

Both models achieve strong overall performance in the baseline configuration. R3M achieves a mean success rate of 50.2%, 78.8%, and 89.3% at 5, 10, and 25 demonstrations respectively, while VIP achieves 52.0%, 77.9%, and 86.2% under the same conditions. The two models are broadly comparable across all conditions, which is consistent with the findings reported in both original papers [1], [2].

Task difficulty varies substantially across the five tasks. Button Press and Drawer Open saturate quickly, with both models reaching near-perfect performance by 10 demonstrations. Hammer and Bin Picking show moderate difficulty, with meaningful improvement as demonstration count increases. Assembly is the most challenging task for both models, exhibiting the highest variance and the largest gap between 5 and 25 demonstrations, which reflects the precision required for peg insertion and the difficulty of the custom inverse kinematics heuristic used to generate demonstrations for this task.

Camera viewpoint has a relatively modest effect on overall performance for both models, as shown in Fig. 3. Both R3M and VIP maintain consistent success rates across the front, side, and top-down camera angles, with no single viewpoint consistently outperforming the others. This finding is broadly consistent with the viewpoint robustness reported in the R3M paper [1], and suggests that within-domain viewpoint variation in simulation is less impactful than the larger domain gap

[1] `.venv/lib/python3.12/site-packages/hydra/conf/__init__.py`

TABLE I: MetaWorld Behavioral Cloning Success Rate (%) for R3M and VIP. Results are averaged over 3 random seeds and 3 camera viewpoints. Mean ± standard deviation are reported.

| | 5 Demos | | 10 Demos | | 25 Demos | |
|---|---|---|---|---|---|---|
| **Task** | **R3M** | **VIP** | **R3M** | **VIP** | **R3M** | **VIP** |
| Button Press | 80.0±17.0 | 71.7±9.4 | 100.0±0.0 | 98.9±2.2 | 100.0±0.0 | 100.0±0.0 |
| Drawer Open | 67.2±13.5 | 76.1±12.9 | 97.8±3.6 | 99.4±1.7 | 100.0±0.0 | 100.0±0.0 |
| Hammer | 41.1±12.2 | 50.0±14.1 | 76.7±11.5 | 82.8±11.5 | 92.2±6.7 | 91.1±7.0 |
| Bin Picking | 30.6±7.3 | 31.7±15.4 | 50.6±20.4 | 60.6±12.9 | 76.7±13.5 | 71.1±23.2 |
| Assembly | 32.2±10.3 | 30.6±11.6 | 68.9±9.3 | 47.8±10.9 | 77.8±24.3 | 68.9±22.0 |
| **Average** | **50.2**±12.0 | **52.0**±12.7 | **78.8**±8.9 | **77.9**±7.8 | **89.3**±8.9 | **86.2**±10.4 |

TABLE II: MetaWorld behavioral cloning success rate (%) for R3M and VIP after fine-tuning the projection head on the DROID exocentric robot manipulation dataset. Results are averaged over 3 random seeds and 3 camera viewpoints. Mean ± standard deviation are reported.

| | 5 Demos | | 10 Demos | | 25 Demos | |
|---|---|---|---|---|---|---|
| **Task** | **R3M+DROID** | **VIP+DROID** | **R3M+DROID** | **VIP+DROID** | **R3M+DROID** | **VIP+DROID** |
| Button Press | 71.7±10.0 | 73.3±11.2 | 97.8±3.6 | 99.4±1.7 | 98.9±2.2 | 100.0±0.0 |
| Drawer Open | 64.4±21.9 | 74.4±11.6 | 97.8±3.6 | 95.6±6.8 | 98.9±3.3 | 100.0±0.0 |
| Hammer | 35.6±9.2 | 38.3±15.0 | 72.2±17.9 | 75.6±17.0 | 90.6±9.5 | 86.1±18.7 |
| Bin Picking | 37.2±17.5 | 26.7±12.2 | 50.0±19.5 | 40.6±20.8 | 60.0±15.4 | 51.7±18.2 |
| Assembly | 18.3±10.9 | 17.8±10.3 | 12.2±10.9 | 16.7±16.4 | 18.9±10.5 | 26.1±16.5 |
| **Average** | **45.4**±13.9 | **46.1**±12.1 | **66.0**±11.1 | **65.6**±12.6 | **73.4**±8.2 | **72.8**±10.7 |

TABLE III: MetaWorld behavioral cloning success rate (%) for R3M and VIP after fine-tuning with the ResNet-50 backbone frozen (projection head only) on the DROID dataset. Results are averaged over 3 random seeds and 3 camera viewpoints. Mean ± standard deviation are reported.

| | 5 Demos | | 10 Demos | | 25 Demos | |
|---|---|---|---|---|---|---|
| **Task** | **R3M+DROID** | **VIP+DROID** | **R3M+DROID** | **VIP+DROID** | **R3M+DROID** | **VIP+DROID** |
| Button Press | 72.2±13.5 | 67.8±8.3 | 95.6±6.8 | 98.9±2.2 | 100.0±0.0 | 100.0±0.0 |
| Drawer Open | 56.1±15.0 | 75.0±14.1 | 97.2±5.1 | 97.8±4.4 | 100.0±0.0 | 100.0±0.0 |
| Hammer | 39.4±13.8 | 38.9±5.5 | 73.3±15.0 | 70.0±12.2 | 85.8±8.0 | 82.8±22.4 |
| Bin Picking | 33.3±18.2 | 21.1±14.3 | 46.7±16.8 | 41.1±17.1 | 61.7±13.3 | 58.9±21.0 |
| Assembly | 13.3±6.1 | 21.7±8.7 | 16.1±16.2 | 16.1±11.1 | 32.5±17.2 | 32.8±20.3 |
| **Average** | **42.9**±13.3 | **44.9**±10.2 | **65.8**±12.0 | **64.8**±9.4 | **76.0**±7.7 | **74.9**±12.7 |

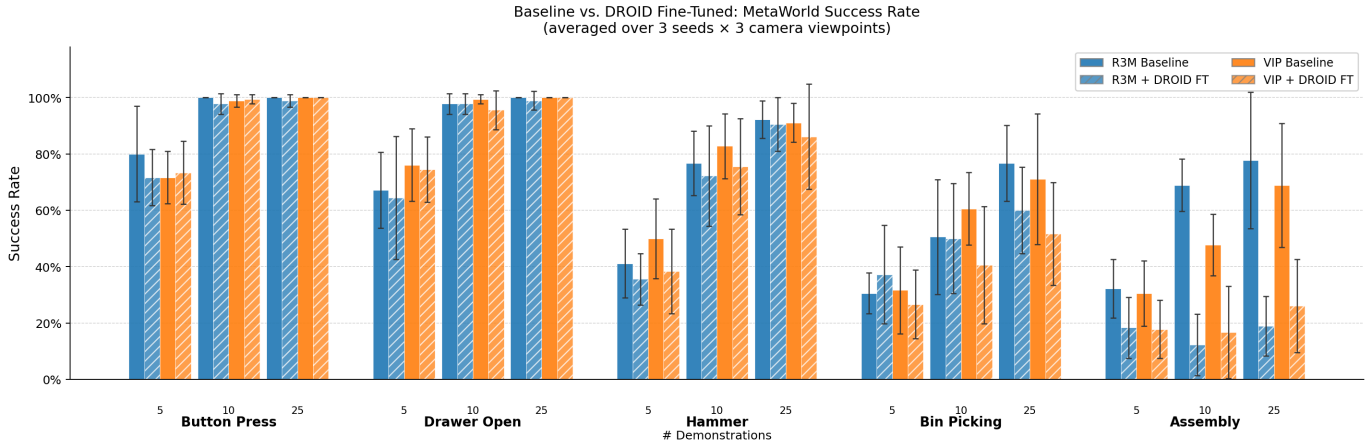



Fig. 2: MetaWorld success rate for R3M and VIP. Baseline and DROID fine-tuned results are shown across all five tasks and three demonstration dataset sizes (5, 10, and 25 episodes). Results are averaged over 3 random seeds and 3 camera viewpoints. Error bars denote standard deviation.

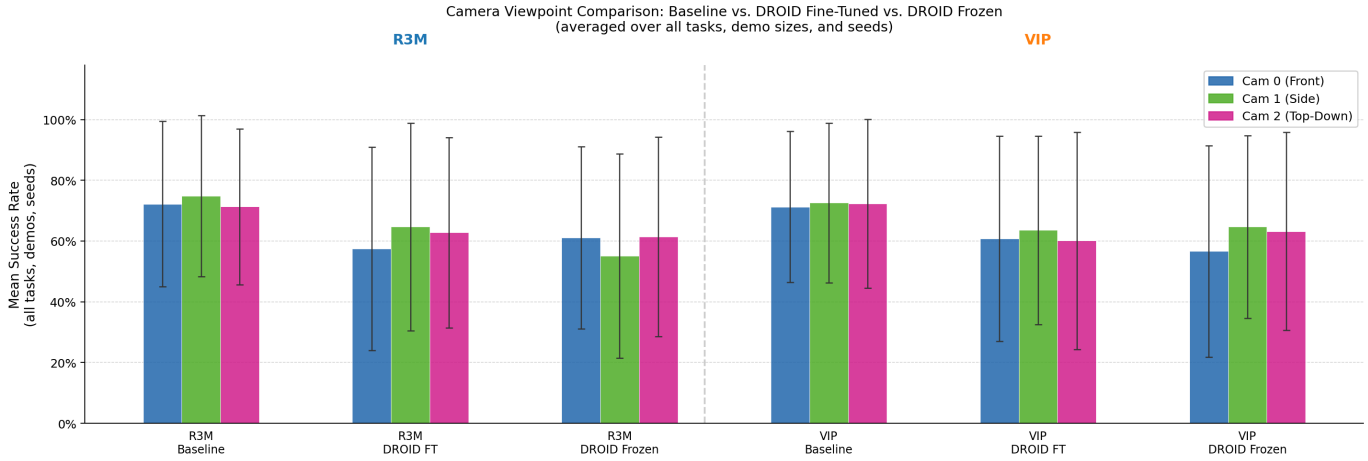
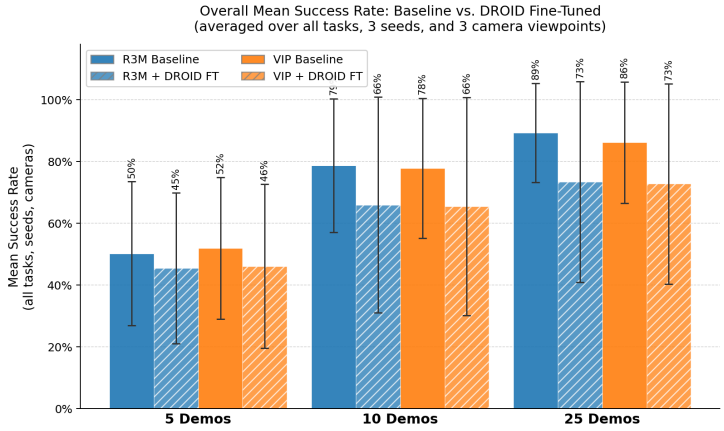
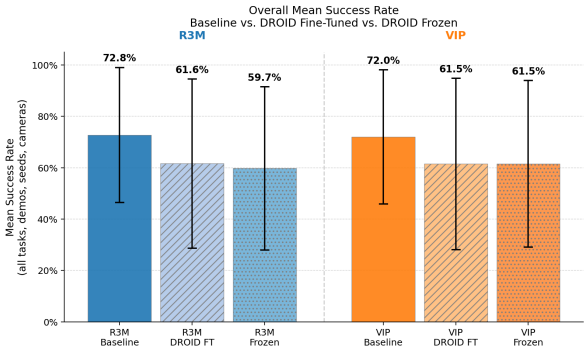



Fig. 3: MetaWorld success rate for R3M and VIP across various camera angles.




(a) Average success rate per number of demonstrations, collapsed over all tasks.




(b) Overall mean success rate across all tasks, demo sizes, seeds, and camera viewpoints.

Fig. 4: Aggregated performance comparison between R3M and VIP on the MetaWorld benchmark. Both models achieve comparable overall performance, however, significant reduction in success is observed after fine-tuning with DROID.

between egocentric pretraining data and third-person robot cameras.

### B. *Effect of DROID Fine-Tuning*

We next evaluate the effect of fine-tuning both encoders on a subset of the DROID dataset [4]. Two fine-tuning strategies were tested: full backbone fine-tuning, in which all ResNet-50 parameters were updated, and frozen backbone fine-tuning, in which only the projection head was updated while the convolutional backbone remained frozen. Results for both strategies are reported in Table II and Table III respectively, and Fig. 4b shows the overall comparison across all four conditions.

Contrary to our hypothesis, DROID fine-tuning consistently degraded performance across both models and all task and demonstration conditions regardless of strategy. Under full backbone fine-tuning, R3M dropped from an overall mean of 72.8% to 61.6%, and VIP dropped from 72.0% to 61.5%, a decrease of approximately 11 percentage points for both models. Under frozen backbone fine-tuning, results were nearly identical: R3M reached 61.6% and VIP reached 61.5%. The near-identical outcomes across both strategies are an important finding in themselves: since freezing the backbone eliminates catastrophic forgetting of pretrained features by design, and yet the degradation remained the same, catastrophic forgetting is ruled out as the primary cause of the performance drop.

The degradation was not uniform across tasks. Button Press and Drawer Open, which saturate at high performance in the baseline, were largely unaffected at higher demonstration counts. However, Assembly showed the most severe degradation, dropping from 68.9% to 12.2% for R3M and from 47.8% to 16.7% for VIP at 10 demonstrations, with near-identical drops under the frozen strategy. Bin Picking and Hammer also declined, though less dramatically. The standard deviation of results increased substantially after DROID fine-tuning under both strategies, suggesting that the fine-tuned encoders

produce less consistent representations than the originals.

Given that catastrophic forgetting is ruled out, the remaining hypotheses center on the nature of the DROID data itself. First, Ego4D was recorded from a human first-person perspective, meaning R3M's features may be tuned to human morphology, specifically the appearance of human hands and arms in motion. DROID shows a Franka Panda gripper, which is visually very different from a human hand. Even with the backbone frozen, the projection head fine-tuned on robot gripper imagery may have learned to map these morphologically distinct features into an embedding space that is less useful for the downstream behavioral cloning policy. Second, the fine-tuning subset covers diverse real-world manipulation tasks that are visually and structurally quite different from MetaWorld's simulated tabletop tasks. The domain gap between DROID's real-world environments and MetaWorld's simulation rendering may be as large or larger than the original egocentric-to-exocentric gap the fine-tuning was intended to close. Together, these results suggest that the visual domain and task structure of the fine-tuning data must be closely aligned with the downstream evaluation environment for exocentric fine-tuning to be beneficial.

### C. Limitations

Several limitations of this evaluation should be noted. First, evaluations are restricted to MetaWorld due to difficulties obtaining compatible demonstration data for Franka Kitchen and Adroit, as described in Section IV-G. Second, only approximately 8% of the full DROID dataset was used for fine-tuning due to local storage constraints, which may have limited the quality and diversity of the fine-tuning signal. Third, the originally planned fine-tuning source, Ego-Exo4D, was not used due to data access constraints; its paired egocentric and exocentric recordings of human manipulation activities may have produced different results than the robot-centric DROID data. Fourth, while the frozen backbone experiment ruled out catastrophic forgetting as the primary cause of degradation, the underlying cause – whether morphological mismatch, domain shift from simulation, or both – remains unconfirmed and warrants further investigation. Finally, all evaluations are conducted in simulation, and real-robot transfer is not assessed.

## VI. Conclusion

This paper investigated whether the egocentric pretraining bias of R3M and VIP limits their performance in robot manipulation, and whether fine-tuning on exocentric data can address this gap. We reproduced the behavioral cloning evaluation of both models on five MetaWorld tasks across three demonstration sizes, three camera viewpoints, and three random seeds, confirming that both models achieve strong and comparable baseline performance consistent with their original papers. R3M achieved an overall mean success rate of 72.8% and VIP achieved 72.0%, averaged across all conditions.

We then fine-tuned both encoders on a subset of the DROID dataset using two strategies: full backbone fine-tuning and frozen backbone fine-tuning (projection head only). Both strategies produced nearly identical results, with overall performance dropping to approximately 61.6% for both models. The fact that freezing the backbone did not change the outcome is a meaningful finding: it rules out catastrophic forgetting as the primary driver of degradation and points instead to the nature of the fine-tuning data itself. Morphological mismatch between the human hands depicted in Ego4D and the robot grippers in DROID, and the visual and structural gap between DROID's real-world environments and MetaWorld's simulation, are the most plausible remaining explanations.

Taken together, these findings suggest two conclusions. First, R3M and VIP are already robust to viewpoint variation within the simulation domain, meaning the egocentric pretraining gap may manifest most strongly in real-robot deployment rather than in simulation. Second, improving these representations through exocentric fine-tuning requires careful alignment between the fine-tuning data and the downstream evaluation environment.

Several directions remain open for future work. The originally planned Ego-Exo4D fine-tuning, using paired human activity video from cooking and tool-use scenarios, remains the most directly motivated next step and warrants pursuit once data access is obtained – such data may better preserve the human morphology features that R3M and VIP rely on while still introducing third-person viewpoints. Fine-tuning on a larger portion of DROID or a filtered subset more closely matched to MetaWorld task structure may also help isolate whether the degradation is due to morphological mismatch or domain shift. Partial backbone fine-tuning, unfreezing only the final residual block, represents a middle ground between the two strategies tested here. Finally, evaluating fine-tuned encoders on real robot hardware would more directly test whether the egocentric pretraining bias affects real-world deployment performance.

## Reproducibility Statement

All code, experiment scripts, and result data for this paper are publicly available at https://github.com/DavidRoss85/ego_exo_analysis_26. The repository includes the following components:

- Multitask behavioral cloning evaluation scripts for R3M and VIP on the MetaWorld benchmark, covering all five tasks, three camera viewpoints, and three demonstration dataset sizes (5, 10, 25).
  - `r3m_metaworld_multitask.py`
  - `vip_metaworld_multitask.py`
- Fine-tuning scripts for both encoders on the DROID and Ego-Exo4D
  - `r3m_droid_finetune.py`
  - `vip_droid_finetune.py`
  - `r3m_egoexo4d_finetune.py`
  - `vip_egoexo4d_finetune.py`
- Raw CSV result files for all reported experiments, including baseline MetaWorld evaluations and DROID fine-

tuning results across all seeds, camera viewpoints, and demonstration conditions.

Pre-trained R3M and VIP model weights are available from their respective original repositories [1], [2]. Ego-Exo4D data requires a separate access request through the dataset's official portal [5]; DROID is similarly subject to its own terms of access [4]. Key hyperparameters are documented in Section IV: behavioral cloning policies are trained for 20,000 gradient steps with a batch size of 32 and learning rate of $10^{-3}$. Three random seeds are used throughout; all seeds are fixed and recorded in the result files. Instructions for environment setup and experiment execution are provided in the repository README.

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